

Principles of Economics

Class 11: Unemployment and Fiscal Policy

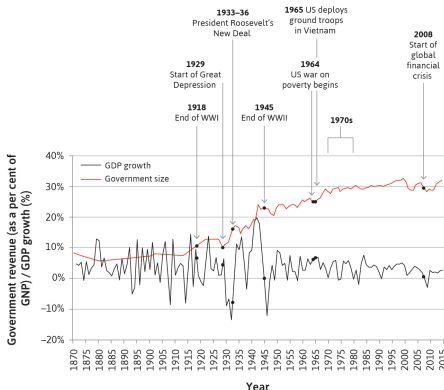
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Introduction

- ▶ The chart shows:
 - ▶ US GDP and the size of government measured as $\frac{\text{Government Revenue}}{\text{GNP}}$ (GNP: gross national product $\approx$ GDP)
- ▶ What is striking about the chart?



- ▶ Last class: Aggregate Demand (**GDP**) **fluctuates** with Consumption and Investment (Macro, Unit 3)
- ▶ **Fluctuations carry large economic and social costs**: which forces smooth them? We also saw:
 - ▶ **Consumption** matters, but credit constraints limit it.
 - ▶ **Investment** is more volatile than GDP.
- ▶ This class: how **governments** dampen fluctuations with fiscal policy:
 - ▶ How can the **government stabilise the Economy**?
 - ▶ Why are some **government policies not effective**?
 - ▶ How to model output and unemployment together?
- ▶ **Fiscal policy**: taxes and public spending.

- ▶ **Components of aggregate demand (AD):**
 - ▶ Private Consumption
 - ▶ Private Investment
 - ▶ Government
 - ▶ Net Exports
- ▶ Already seen: they measure economic activity and GDP (expenditure side)
- ▶ Now: how the government uses them to stabilise the economy.

The (private) Consumption Function

The (private) Investment Function

The Basic Multiplier Model

The Multiplier with Government and the Foreign Sector

The Multiplier and Fiscal Policy

Public Finances

Reflections on the Multiplier

Conclusion

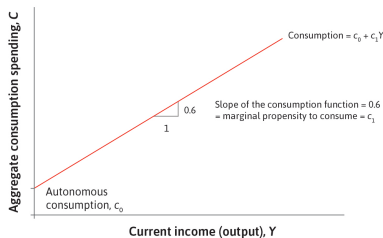
1. The (private) Consumption Function

- ▶ Consumption smoothing implies two parts (individual and aggregate):

$$C = c_0 + c_1 Y$$

where

- ▶ c_0 : **Autonomous Consumption** (independent of current income)
- ▶ $c_1 Y$: **Income-dependent consumption** (proportional to current income)
- ▶ $c_1 (0 < c_1 < 1)$ is the **marginal propensity to consume (MPC)**



What determines autonomous consumption and the marginal propensity to consume?

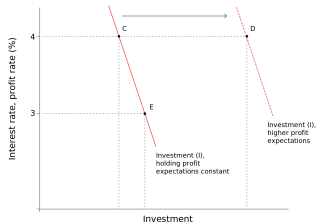
- ▶ **The ability to smooth shocks determines the MPC:**
 - ▶ **Poor households:** react more to income changes, credit-constrained [**high MPC**]
 - ▶ **Rich households:** react less to current income [**small MPC**]
- ▶ **Expectations on future income and wealth determine autonomous consumption:**
 - ▶ Housing, if owned by the household:
 - ▶ If it was bought with a **loan**:
 - ▶ **The outstanding value of the loan is discounted;**
 - ▶ Savings and other investments are financial wealth;
 - ▶ Uncertainty raises **precautionary saving** → autonomous consumption falls.

2. The (private) Investment Function

- ▶ **Investment is autonomous:** it does not depend on income
- ▶ The **level of investment** depends on:
 - ▶ The owner's discount rate (ρ)
 - ▶ The interest rate (r)
 - ▶ The net rate of return on investment (Π)
- ▶ **Investment increases** with:
 - ▶ A lower interest rate
 - ▶ A better business environment
 - ▶ Higher expected profits
e.g. lower risk of expropriation.

The (private) investment function

- ▶ **Aggregate investment function:** investment against the interest rate and expected profits.
- ▶ **Investment decreases** with the interest rate (r):
 - ▶ The investment curve is downward-sloping
- ▶ **Higher expected profits** → the investment function **shifts right**:
 - ▶ Same interest rate ($r = 4\%$), higher investment.
 - ▶ Possible causes:
Expectations, a fall in input prices (energy costs/wages), technological improvements, lower taxes, etc.
- ▶ In the data: **less sensitive** to the interest rate than to uncertainty and expectations.
- ▶ **The interest rate** acts indirectly, via durable-goods consumption



3. The Basic Multiplier Model

The basic multiplier model

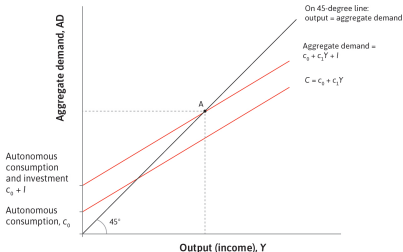
Equilibrium in the goods market (I)

- Basic model of **aggregate demand: consumption and investment:**

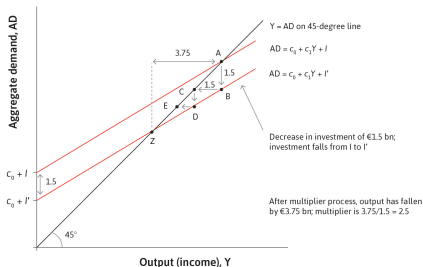
$$DA = C + I = c_0 + c_1 Y + I = c_0 + I + c_1 Y$$

where

- $c_0 + I$: **Intercept with the vertical axis:** autonomous consumption and investment
- On the **45° line**: output = aggregate demand
- Equilibrium**: AD cuts the 45° line (point A):
 $Y = DA \iff \frac{1}{1-c_1}(c_0 + I)$
- Note that:
 - The slope of the 45° line is 1;
 - The slope of AD is c_1 , because $MPC = c_1 < 1$.
- Goods market equilibrium: $Y = AD$ (output = demand).



- Fall in investment:
 - fall in aggregate demand
 - lower output and income
 - ...
 - new equilibrium (Z)



- **Multiplier effect:** Investment falls €1.5bn and output falls €3.75bn (multiplier=2.5)

When is the multiplier larger?

The basic model of aggregate demand and the multiplier

Equilibrium in the goods market (I): the multiplier effect

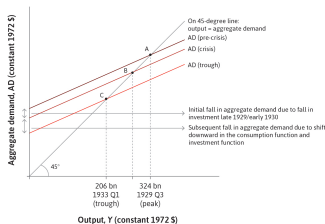
- ▶ **Multiplier effect:** total change in output > initial change in demand:
- ▶ Cause: the **circular flow of expenditure, income and output**.

$$Y = DA \iff \frac{1}{1-c_1}(c_0 + I)$$
- ▶ GDP (Y) and AD move together; the **multiplier gives the relative magnitude**:
 - ▶ Multiplier > 1: more than proportional change in GDP.
 - ▶ Multiplier = 1: proportional change in GDP.
 - ▶ Multiplier < 1: less than proportional change in GDP.
- ▶ What determines **the size of the multiplier in the basic model**?
 - ▶ The **slope of AD**: the **marginal propensity to consume**.
 - ▶ **E.g., credit constraints and preferences for consumption smoothing**.
- ▶ **Changes in AD**: from **changes in the MPC** or in the **autonomous components** [consumption and investment]

The basic model of aggregate demand and the multiplier

Equilibrium in the goods market (I): the crisis of 1929

- ▶ This simple AD model applied to the **1929 crisis**:
- ▶ **Before the crisis**: GDP of 324 bn (Q3, 1929):
 - ▶ Where the brown AD crosses the 45° line [**point A**].



- ▶ **1929 stock market crash**: investment drops and aggregate demand falls;
 - ▶ New equilibrium with lower output/income [**point B**];
 - ▶ The fall in GDP is more than proportional: multiplier > 1
- ▶ With the **fall in incomes**, AD fell even further:
 - ▶ A fall in consumption and a further fall in investment:
 - ▶ Causes: uncertainty after the crash, pessimism, banking crisis, credit collapse.
 - ▶ New equilibrium output of 206 billion [**point C**], multiplier > 1 .

4. The Multiplier with Government and the Foreign Sector

- ▶ Now add **Government** (G) and **foreign trade** (NX) to the AD model:
 $DA = C + I + G + NX$
- ▶ The **Government** influences AD in three ways:
 - ▶ **Government Spending** (G): autonomous (exogenous component), shifts AD upwards;
 - ▶ **Consumption** (C): income taxes (t) cut disposable income, and with it consumption.
 - ▶ $C = c_0 + c_1(1 - t)Y$
 - ▶ **Investment** (I): depends on the interest rate and other factors (corporate taxes, business environment)
 - ▶ A high interest rate or tax rate reduces autonomous investment.
- ▶ **Foreign trade** influences AD in two ways:
 - ▶ **Exports** (X): considered exogenous and shift AD upwards;
 - ▶ **Imports** (M): assumed to be proportional to income
 - ▶ $M = mY$
 - ▶ **Marginal propensity to import** (m): share of each extra unit of income spent on imports.

- ▶ With these components, the **AD model becomes**:

$$AD = C + I + G + NX = c_0 + c_1(1 - t)Y + I + G + X - mY$$

$$\Rightarrow AD = c_0 + I + G + X + [c_1(1 - t) - m]Y$$

where

- ▶ c_0, I, G, X : **Autonomous Components** (intercept with the vertical axis).
- ▶ $c_1(1 - t)Y, mY$: **Endogenous Components** (slope and multiplier).
- ▶ Solving $AD = Y$ for Y , we have: $Y = \frac{1}{1 - [c_1(1 - t) - m]}(c_0 + I + G + X)$
 - ▶ If c_1 is small, and t and m are high:
 - ▶ AD is flatter;
 - ▶ Smaller multiplier.
- ▶ **Saving, taxes and imports are leakages from the circular flow of income: they reduce the size of the multiplier:**
 - ▶ c : what share of income goes to consumption and saving;
 - ▶ t : what share of income goes directly to the government as taxes;
 - ▶ m : what share of income is used to buy goods from abroad

- ▶ **The government stabilises economic fluctuations** in several ways:
 - ▶ **Government spending** is large and exogenous
 - ▶ **A higher tax rate** reduces the multiplier [it reduces disposable income and thus the role of consumption]
 - ▶ **Unemployment insurance** helps smooth consumption
 - ▶ Intervention through **fiscal policy**.
- ▶ **Unemployment benefits and the proportional tax rate** are the **automatic stabilisers**;
- ▶ **They automatically offset the expansion or contraction** of the economy;
 - ▶ The **proportional tax rate** cuts induced consumption more in booms;
 - ▶ **Unemployment benefits** cushion the fall in incomes in a recession.

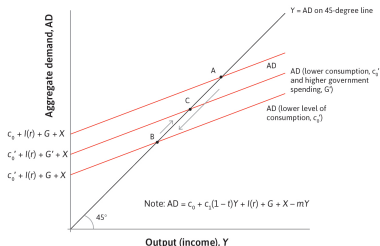
5. The Multiplier and Fiscal Policy

Note: the Paradox of thrift

- ▶ In a **recession**, a **family** worried about its wealth **cuts spending and saves more**;
- ▶ But in the economy as a whole, **spending and earnings go hand in hand**:
 - ▶ If consumption falls, so does firms' output;
 - ▶ Lower investment and employment follow;
 - ▶ And a further fall in household income and consumption.
 - * **Paradox of thrift**: trying to save more in the aggregate lowers aggregate income. So the best reaction to a fall in income may be to spend more.
- ▶ The paradox of thrift is an example of the **fallacy of composition**:
 - ▶ What is true for a part of the economy (a single household) is not necessarily true for the whole economy.

Fiscal stimulus

- ▶ In a **recession**, fiscal stimulus offsets the fall in private-sector AD:
- ▶ This is a **discretionary change in fiscal policy**, a countercyclical policy.
 - ▶ **Tax cut** ($\downarrow t$): to encourage private spending
– Indirect effect.
- ▶ Start with the **economy at point A**:
 - ▶ **Autonomous consumption falls** ($c_0 \rightarrow c_0'$)
 - ▶ Equilibrium output **falls** to point B.
- ▶ If the government **increases spending** ($G \rightarrow G'$), AD rises:
 - ▶ Equilibrium **output/income increases** to C
 - ▶ $Y_B < Y_C < Y_A$
- ▶ **Through the multiplier**: government spending moves output **more than proportionally**.



Financing a fiscal stimulus

- ▶ **Fiscal stimulus packages** mean more spending or less revenue, and that **strains the budget**;
 - ▶ Government budget ...
 - ▶ Surplus: $\text{Revenue} > \text{Spending}$
 - ▶ Balance: $\text{Revenue} = \text{Spending}$
 - ▶ Deficit: $\text{Revenue} < \text{Spending}$
- ▶ In our **simplified model**, the government:
 - ▶ Revenue: income tax collection (T)
 - ▶ Spending: government spending (G)
 - ▶ Budget balance = $T - G$
 - ▶ If G and T increase at the same time, the stimulus is small or nonexistent.
- ▶ So stimulus **generates a deficit**, financed with government debt:
 - ▶ If the deficit is temporary, it is not problematic;
 - ▶ **Persistent deficits** will generate an increase in the level of government debt;
 - It leaves the government less room in future crises.

Austerity policy

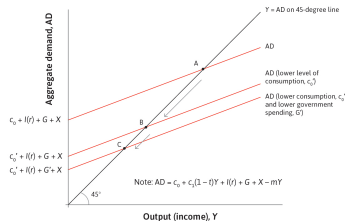
- ▶ In a **recession** the public deficit rises through the **automatic stabilisers**:
 - ▶ **Tax revenues will decrease** because output is lower;
 - ▶ Unemployment benefits **rise**.
- ▶ If the Government is subject to high levels of public debt:
 - ▶ Stimulus policies may not be possible;
 - ▶ It may be unsustainable if the cost of the **debt is too high**.

- ▶ The government may then be forced into **austerity** [equilibrium point C]:

- ▶ $\downarrow G$ or $\uparrow t$;
- ▶ This will reduce AD further ($Y_C < Y_B < Y_A$).

- ▶ In this context, it is a **procyclical policy**.

- ▶ **Through the multiplier**, the fall in government spending cuts output **more than proportionally**.

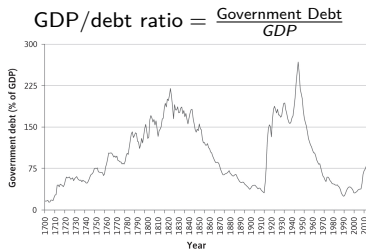


6. Public Finances

- ▶ **Two concepts of deficit:**
 - ▶ **Primary budget deficit:** Government spending - Taxes (G-T) **excluding interest spending** on the debt.
 - ▶ **Total budget deficit:** Government spending - Revenues + **Interest spending**
- ▶ **Total government debt:** all the bonds sold over time to finance budget deficits:
 - ▶ Bonds can be issued with different maturities: short (< 5 years) or long term (≥ 5 years);
 - ▶ Matured bonds are debts that have already been paid.
 - ▶ The bond price is the interest rate [a measure of risk] that investors are willing to pay to hold the bond.
- ▶ A large debt (debt/GDP ratio) is a problem if default risk is perceived:
 - ▶ No moment requires paying off the whole stock: the government issues new bonds.
 - ▶ An ever-increasing debt/GDP ratio is unsustainable, but there is no rule that sets a specific limit.
- ▶ **Sovereign debt crisis:** government bonds come to be seen as (very) risky.

Debt as a percentage of GDP

- ▶ The chart: the UK Debt/GDP ratio since 1700:
- ▶ We measure **indebtedness relative to the size of the economy** (debt/GDP ratio):
 - ▶ A smaller country, like Luxembourg, has a smaller debt than Germany.
 - ▶ What matters is its relative size.
- ▶ **Indebtedness measured** by the debt/GDP ratio falls if:
 - ▶ **The primary budget balance is positive** (so it can be used to pay down debt).
 - ▶ **GDP grows faster** than government debt.
 - ▶ **Inflation is high** (it lowers the real value of the debt).
- ▶ **Bonds are paid at their face value.** If you borrowed \$1000 in 1980 and have to pay it back in 2020, the money has lost value due to inflation, but you still pay \$1000.



7. Reflections on the Multiplier

Aggregate demand: feedback

The private sector, government actions and (equilibrium) output

- ▶ Both the **private sector** and the **government** can help **stabilise output**.
- ▶ Crucial to **avoid excessive booms and deep recessions**;
- ▶ The **government** usually has a **stabilising role** but there are **policy mistakes**, such as:
 - ▶ Unemployment benefits that are too strict;
 - ▶ Systematic budget deficits.
- ▶ It limits **the tools available** to stabilise output [for example, austerity in a recession].

	DAMPENING MECHANISMS OFFSET SHOCKS (STABILISING)	AMPLIFYING MECHANISMS REINFORCE SHOCKS (MAY BE DESTABILISING)
PRIVATE SECTOR DECISIONS	Consumption smoothing	Credit constraints limit consumption smoothing Rising value of collateral (house prices) can increase wealth above the target level and raise consumption Rising capacity utilisation in a boom encourages investment spending, adding to the boom
GOVERNMENT AND CENTRAL BANK DECISIONS	Automatic stabilisers (e.g. unemployment benefit) Stabilisation policy (fiscal or monetary)	Policy mistakes, such as limiting the scope of automatic stabilisers in a recession or running deficits during low demand periods, while not running surpluses during booms

Aggregate demand and unemployment

The labour market and the multiplier models

► Aggregate unemployment from two perspectives:

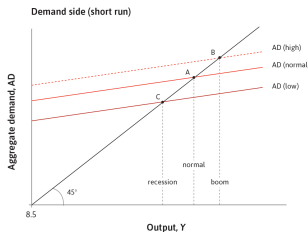
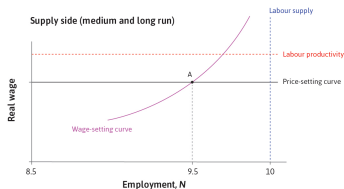
► In the **medium run**:

- Both wages and prices are flexible.
- But the capital stock, technology and institutions are fixed.
- The **labour market** model, from the supply side [classes 8 and 9]

► In the **short run**:

- All variables are fixed.
- The **multiplier** model → from the demand side.

- **Fluctuations in AD** around the labour-market equilibrium cause **cyclical unemployment** - neither the wage-setting nor the price-setting curves change in the short run!



Calculating the size of the multiplier

- ▶ In our AD model the **multiplier depended only** on the slope of AD:
 - ▶ Marginal Propensity to consume (c);
 - ▶ Marginal Propensity to import (m);
 - ▶ Income tax rate (t)
- ▶ In reality many **other factors** matter:
 - ▶ **Capacity utilisation rate** [phase of the business cycle]:
 - ▶ With fully employed resources [boom], increases in government spending can **crowd out private spending**;
 - ▶ Private sector **expectations**:
 - ▶ **The multiplier can be negative** if a bigger deficit erodes the confidence of consumers and firms:
 - ▶ Deficits can be financed with debt, but **debt means higher future taxes and lower future spending**.
- ▶ A large literature **investigates the sign and the size of the fiscal multiplier**:
 - ▶ Question: What is the impact of fiscal policy on output?

The multiplier in real life

Calculating the size of the multiplier

- ▶ The main obstacle is the usual one in economics, **endogeneity**:
 - ▶ Suppose **government spending and output are positively correlated**;
 - ▶ Can we infer that **changes in government spending lead to changes in output**?
 - ▶ **Not necessarily**: the change in government spending may be the result of changes in output, **reverse causality**.
- ▶ **Acconcia, Corsetti and Simonelli (2014)** address exactly this:
- ▶ In Italy, local governments infiltrated by the mafia **are dismissed** and replaced by technocrats:
 - ▶ 199 municipal dismissals unrelated to local economic activity;
- ▶ After the dismissal, **local governments' spending decreases**:
 - ▶ There is no reverse causality: the dismissal is unrelated to the local economy.
- ▶ Even so, a 1% fall in government spending leads to a 1.5% fall in local GDP: **A positive relationship between government spending and GDP!**



8. Conclusion

Section by section (Macro, Units 3 and 5)

1. **The consumption function** → $C = c_0 + c_1 Y$: the MPC says how much of an extra euro is consumed, and it is higher for credit-constrained households.
2. **The investment function** → autonomous and decreasing in the interest rate; it shifts right with expected profits.
3. **The basic multiplier model** → equilibrium where AD cuts the 45° line; any shock is amplified, as in 1929.
4. **Government and the foreign sector** → taxes and imports are leakages: they flatten AD and shrink the multiplier.
5. **The multiplier and fiscal policy** → stimulus in a recession, austerity as a pro-cyclical policy, and the paradox of thrift.
6. **Public finances** → primary and total deficit, debt/GDP: debt is a problem when default risk is perceived.
7. **The multiplier in real life** → it depends on spare capacity and expectations; Acconcia et al. (2014) estimate 1.5 by solving the endogeneity problem.

Key concepts: aggregate demand, autonomous consumption, marginal propensity to consume, autonomous investment, 45° line, multiplier, leakages, marginal propensity to import, automatic stabilisers, fiscal stimulus, austerity, paradox of thrift, fallacy of composition, primary and total deficit, debt/GDP ratio, sovereign debt crisis, cyclical unemployment, endogeneity and reverse causality.